Class 6 Science – Chapter 5: Measurement of Length and Motion

नीचे आपके दिए गए अध्याय के आधार पर सभी दिखाई देने वाले textbook questions, activities और quick checks के उत्तर simple English में दिए गए हैं। इसके बाद प्रत्येक topic से महत्वपूर्ण और संभावित परीक्षा-प्रश्नों के उत्तर भी दिए गए हैं।

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1. Places in My Neighbourhood (Page 62)

इस गतिविधि में अपने घर से स्कूल, किराना दुकान, हेयर सैलून और पार्क की दूरी तथा वहाँ जाने के साधन लिखने हैं। इनके उत्तर हर विद्यार्थी के लिए अलग होंगे।

Q1. Which place is the nearest to your house?

Answer: The nearest place to my house is the grocery store.
(Write the place that is actually nearest to your house.)

Q2. Which place is the farthest from your house?

Answer: The school is the farthest from my house.
(Change the answer according to your actual distances.)

Q3. What is the common unit of measurement of these distances?

Answer: The common units are metres and kilometres.

Q4. Do you walk to any of these places?

Answer: Yes, I walk to the nearby grocery store and park.

2. Course of Development of Transport (Pages 63–64)

Q1. How did people travel in ancient times?

Answer: In ancient times, people walked and carried loads on their heads or backs. They also used animals such as horses, bullocks, donkeys, camels and elephants.

Q2. Which invention brought a revolution in transport?

Answer: The invention of the wheel brought a revolution in transport.

Q3. How did the invention of the wheel help people?

Answer: It made travelling easier, faster and more comfortable. Animals could pull carts with wheels.

Q4. How did rail transport develop?

Answer: Rail transport developed with the invention of the steam engine. Steam engines pulled railway carriages and wagons.

Q5. How did water transport develop?

Answer: People first used wooden planks and logs to travel on water. Later, they made boats and developed motorised boats and ships.

Q6. When were aeroplanes developed?

Answer: Aeroplanes were developed in the early 1900s.

Q7. Name the three main modes of transport.

Answer: The three main modes of transport are land transport, water transport and air transport.

Q8. Give two examples of each mode of transport.

Answer:

- Land transport: Bus and train.
- Water transport: Boat and ship.
- Air transport: Aeroplane and helicopter.

Q9. Why do we need to measure the distance between two places?

Answer: We need to measure the distance to estimate how far a place is and to choose a suitable mode of transport.

Q10. Which invention is shown as important in the development of transport?

Answer: The wheel is shown as an important invention in the development of transport.

3. Measurement and Its Need (Page 64)

Q1. What is measurement?

Answer: Measurement is the process of finding the size or amount of an unknown quantity, such as length, distance, weight or time.

Q2. What is a unit of measurement?

Answer: A unit of measurement is a standard value used to measure a quantity.

Q3. Why do we need measurement?

Answer: We need measurement to know the exact length, distance, weight or time. For example, a tailor measures cloth before stitching clothes.

Q4. Why should we use standard units?

Answer: Standard units give uniform and reliable measurements. They help people obtain the same result when measuring the same object.

4. Non-standard Units of Measurement (Pages 64–65)

Q1. What are non-standard units of measurement?

Answer: Non-standard units are units that do not have a fixed or uniform value. Examples are handspan, cubit, arm length and footstep.

Q2. What is a handspan?

Answer: A handspan is the distance between the tip of the thumb and the tip of the little finger when the hand is stretched.

Q3. What is a cubit?

Answer: A cubit is the distance between the tip of the middle finger and the elbow.

Q4. What is arm length?

Answer: Arm length is the distance from the shoulder to the tip of the middle finger.

Q5. What is a footstep?

Answer: A footstep is the length of one step taken while walking.

Q6. Why are handspan and footstep not reliable units?

Answer: They differ from person to person. Therefore, they do not give uniform and accurate measurements.

Q7. What do you learn by measuring a table using handspans?

Answer: Different people may get different readings because their handspans are different.

Q8. What is the conclusion of the activity on non-standard units?

Answer: Non-standard units are not precise. We should use standard units for accurate measurement.

5. SI Units (Page 65)

Q1. What is the full form of SI?

Answer: SI stands for International System of Units.

Q2. Why was the SI system introduced?

Answer: It was introduced to provide a common and uniform system of measurement throughout the world.

Q3. What is the SI unit of length?

Answer: The SI unit of length is the metre (m).

Q4. Name four other units used to measure length.

Answer: Kilometre, decimetre, centimetre and millimetre.

Q5. Complete the following conversions.

Unit conversion	**Answer**
1 kilometre (km)	1000 metres (m)
1 metre (m)	10 decimetres (dm)
1 decimetre (dm)	10 centimetres (cm)
1 centimetre (cm)	10 millimetres (mm)

Q6. Convert 5000 metres into kilometres.

Answer:

We know that 1000 m = 1 km.

$$5000 \div 1000 = 5\text{ km}$$

Therefore, 5000 m = 5 km.

Q7. Convert 3 metres into centimetres.

Answer:

1 m = 100 cm

3 m = 3 × 100 = 300 cm.

Q8. Convert 250 centimetres into metres.

Answer:

100 cm = 1 m

250 ÷ 100 = 2.5 m.

Q9. Convert 45 millimetres into centimetres.

Answer:

10 mm = 1 cm

45 ÷ 10 = 4.5 cm.

Q10. What is the SI unit of mass and time?

Answer: The SI unit of mass is the kilogram (kg), and the SI unit of time is the second (s).

6. Measurement of Length (Page 66)

Q1. Which tools are used to measure length?

Answer: A ruler, metre scale, metre rod and measuring tape are used to measure length.

Q2. How should we place a ruler to measure an object?

Answer: We should place the ruler parallel to the object, with its markings close to the object.

Q3. Where should the zero mark of the ruler be placed?

Answer: The zero mark should coincide with one end of the object.

Q4. What should we do if the zero mark of a ruler is broken or unclear?

Answer: We should start from another clear marking and subtract the initial reading from the final reading.

Q5. A pencil starts at the 1 cm mark and ends at the 6.3 cm mark. What is its length?

Answer:

Length = Final reading − Initial reading

= 6.3 cm − 1 cm

= 5.3 cm

Q6. How should we keep our eyes while reading a scale?

Answer: We should keep our eyes vertically above the mark being read. This helps us avoid reading errors.

Q7. What is the correct way to measure the length of a book?

Answer: Place the book along the ruler, keep one end at zero and read the mark at the other end. Keep your eyes directly above the mark.

Q8. What mistake is made if a ruler is placed slantingly?

Answer: A slanting ruler may give an incorrect measurement because it is not properly aligned with the object.

7. Comparing Standard and Non-standard Units (Page 67)

Q1. How can you compare standard and non-standard units?

Answer: Measure the same desk using a handspan and a measuring tape. Compare the readings.

Q2. Which method is more accurate: handspan or measuring tape? Why?

Answer: Measuring tape is more accurate because it uses standard units such as centimetres.

Q3. Why may two students get different measurements using handspans?

Answer: Their handspans may be of different lengths.

8. Measurement of a Curved Line (Page 67)

Q1. Can we measure a curved line directly with a straight ruler?

Answer: No. A straight ruler cannot follow the shape of a curved line.

Q2. How can we measure the length of a curved line?

Answer: We can use a thread. Place the thread along the curved line, mark its length, straighten the thread and measure it with a ruler.

Q3. Why is a thread useful for measuring a curved line?

Answer: A thread is flexible and can follow the curve.

Q4. What are the steps for measuring a curved line with a thread?

Answer:

1. Take a long thread and make a knot at one end.
2. Place it carefully along the curved line.
3. Mark the point where the curve ends.
4. Straighten the thread.
5. Measure the marked length using a ruler.

9. Describing Positions (Page 68)

Q1. What is a reference point?

Answer: A reference point is a fixed object used to describe the position of another object.

Q2. Give an example of a reference point.

Answer: If a car is parked 200 metres from a tree, the tree is the reference point.

Q3. What is the use of kilometre stones?

Answer: Kilometre stones show the distance in kilometres from a particular starting point. They help travellers estimate the distance remaining to their destination.

Q4. Does the position of an object change with respect to a reference point when it is in motion?

Answer: Yes. The position of an object changes with respect to a reference point when the object is in motion.

Q5. Can we describe the position of an object without a reference point?

Answer: It is difficult to describe its position clearly without a reference point.

10. Quick Check – True or False (Page 68)

1. There are three modes of transport—air, water and land.
Answer: True

2. In ancient times, measurements were done using the SI unit system.
Answer: False

3. Handspan is the length between the tip of the thumb and the middle finger.
Answer: False — A handspan is measured between the tips of the thumb and little finger when the hand is stretched.

4. Non-standard units of measurement differ from person to person.
Answer: True

5. The length of a curved line can be measured in the same way as a straight line.
Answer: False — A flexible thread can be used to measure a curved line.

11. Motion and Rest (Page 68)

Q1. What is motion?

Answer: Motion is a change in the position of an object with time with respect to its surroundings or a reference point.

Q2. What is rest?

Answer: An object is at rest if its position does not change with time with respect to its surroundings.

Q3. How can we bring an object into motion?

Answer: We can push, pull or throw an object to bring it into motion.

Q4. How can we stop a moving object?

Answer: We can stop a moving object by applying a force, such as pressing the brakes on a bicycle.

Q5. Why does a ball in motion stop after some time?

Answer: Friction between the ball and the ground, along with air resistance, slows down the ball until it stops.

Q6. What describes the state of motion of an object?

Answer: The speed and direction of motion describe the state of motion of an object.

Q7. When is an object considered to be at rest?

Answer: An object is considered to be at rest when its position does not change with time with respect to its surroundings.

12. Types of Motion (Pages 69–70)

A. Translational Motion

Q1. What is translational motion?

Answer: Translational motion occurs when all parts of an object move from one place to another.

Q2. What are the two types of translational motion described in the chapter?

Answer: They are rectilinear motion and curvilinear motion.

B. Rectilinear or Linear Motion

Q1. What is rectilinear motion?

Answer: When an object moves along a straight line, its motion is called rectilinear or linear motion.

Q2. Give four examples of rectilinear motion.

Answer:

- A car moving along a straight road.
- A bus moving along a straight road.
- A fruit falling from a tree.
- A striker moving towards a coin on a carrom board.

C. Curvilinear Motion

Q1. What is curvilinear motion?

Answer: When an object moves along a curved path, its motion is called curvilinear motion.

Q2. Give three examples of curvilinear motion.

Answer:

- A train moving on a curved track.
- A basketball thrown towards a basket.
- A paper aeroplane moving along a curved path.

Q3. What is random motion?

Answer: Random motion is motion in which an object does not follow a fixed path.

Q4. Give two examples of random motion.

Answer: A butterfly flying in a garden and a bee moving over flowers.

D. Rotational Motion

Q1. What is rotational motion?

Answer: When an object rotates around its own axis or a fixed point, it is said to be in rotational motion.

Q2. Give three examples of rotational motion.

Answer:

- A rotating fan.
- The hands of a clock.
- A spinning wheel.

Q3. What is circular motion?

Answer: Circular motion occurs when an object moves along a circular path around a fixed centre or point.

Q4. Give an example of circular motion.

Answer: A point on a rotating wheel moves in a circular path around the wheel's centre.

E. Oscillatory Motion

Q1. What is oscillatory motion?

Answer: The repeated to-and-fro motion of an object about its position of rest is called oscillatory motion.

Q2. Give three examples of oscillatory motion.

Answer:

- A swinging pendulum.
- A swing moving to and fro.
- A needle of a sewing machine moving up and down.

F. Vibratory Motion

Q1. What is vibratory motion?

Answer: Rapid to-and-fro motion of an object about its position of rest is called vibratory motion.

Q2. Give two examples of vibratory motion.

Answer:

- The strings of a guitar after being plucked.
- The surface of a drum when it is played.

Q3. How is sound produced?

Answer: Sound is produced by vibrations.

Q4. What happens when vibrations become faster?

Answer: Faster vibrations generally produce a shriller sound.

G. Periodic Motion

Q1. What is periodic motion?

Answer: Motion that repeats itself after fixed intervals of time is called periodic motion.

Q2. Give three examples of periodic motion.

Answer:

- The movement of a clock's hands.
- The revolution of the Earth around the Sun.
- The revolution of the Moon around the Earth.

H. Non-periodic Motion

Q1. What is non-periodic motion?

Answer: Motion that does not repeat itself at regular intervals of time is called non-periodic motion.

Q2. Give two examples of non-periodic motion.

Answer:

- The changing pace of a heartbeat during different activities.
- The movement of our arm while walking.

I. More Than One Type of Motion (Page 70)

Q1. What type of motion is shown by a moving automobile?

Answer: A car moving on a straight road shows linear motion, while its wheels show rotational motion.

Q2. What types of motion are present in a sewing machine?

Answer: The needle moves in an oscillatory or to-and-fro manner, while the wheel rotates.

Q3. What types of motion does the Earth show?

Answer: The Earth rotates on its axis and revolves around the Sun.

Q4. What types of motion are involved in drawing water from a well using a pulley?

Answer: The pulley shows circular or rotational motion, while the bucket moves up or down in linear motion.

Q5. Can an object show more than one type of motion at the same time?

Answer: Yes. For example, a moving car shows linear motion, while its wheels show rotational motion.

13. Additional Important Questions and Answers

ये प्रश्न अध्याय के सभी मुख्य topics को दोहराने और परीक्षा की तैयारी के लिए उपयोगी हैं।

A. Transport and Its Development

Q1. Why do we use different modes of transport?

Answer: We use different modes of transport according to the distance, time and place we need to travel to.

Q2. Which mode of transport is suitable for travelling across oceans?

Answer: Water transport, such as ships, is suitable for travelling across oceans.

Q3. Which mode of transport is generally the fastest for long-distance travel?

Answer: Air transport is generally the fastest for long-distance travel.

Q4. Name two vehicles that use wheels.

Answer: A bicycle and a car use wheels.

B. Measurement and Units

Q1. What is the difference between standard and non-standard units?

Answer: Standard units have fixed values, while non-standard units may differ from person to person.

Q2. Why is a metre scale better than a handspan for measuring a table?

Answer: A metre scale uses standard units and gives more uniform measurements.

Q3. Which unit would you use to measure the following?

- Distance between two cities: Kilometre
- Length of a pencil: Centimetre
- Thickness of a coin: Millimetre
- Length of a classroom: Metre

Q4. Convert 2.5 km into metres.

Answer: $2.5 \times 1000 = 2500$ m.

Q5. Convert 4 m into millimetres.

Answer: 1 m = 1000 mm, so 4 m = 4000 mm.

Q6. Convert 720 cm into metres.

Answer: $720 \div 100 = \mathbf{7.2}$ m.

Q7. Convert 8 cm into millimetres.

Answer: $8 \times 10 = \mathbf{80}$ mm.

Q8. A table is 2 m long. Express its length in centimetres.

Answer: $2 \times 100 = \mathbf{200}$ cm.

C. Correct Measurement

Q1. What precautions should we take while measuring length with a ruler?

Answer:

1. Place the ruler parallel to the object.
2. Align the zero mark with one end.
3. Keep the ruler close to the object.
4. Read the scale with your eyes vertically above the mark.
5. If the zero mark is damaged, subtract the initial reading from the final reading.

Q2. Why should the ruler be placed parallel to the object?

Answer: It helps us measure the actual length of the object accurately.

Q3. An object begins at 2 cm and ends at 9 cm on a ruler. Find its length.

Answer:

Length = Final reading − Initial reading

= 9 cm − 2 cm = 7 cm.

Q4. How can we measure the length of a curved path?

Answer: We can place a thread along the curved path, mark its endpoints and measure the straightened thread with a ruler.

D. Position, Motion and Rest

Q1. Why is a reference point important?

Answer: It helps us describe the position of an object clearly.

Q2. Can the same object be at rest and in motion at the same time?

Answer: Yes, with respect to different reference points. For example, a child sitting in a moving bus is at rest relative to the bus but in motion relative to the road.

Q3. Is a book lying on a table in motion or at rest?

Answer: It is at rest relative to the table and the room.

Q4. Is flowing river water in motion?

Answer: Yes. Its position changes with time relative to the riverbank.

E. Types of Motion

Q1. What is the difference between linear and curvilinear motion?

Answer: Linear motion takes place along a straight line, while curvilinear motion takes place along a curved path.

Q2. Differentiate between rotational and oscillatory motion.

Answer: In rotational motion, an object turns around an axis. In oscillatory motion, an object moves to and fro about its position of rest.

Q3. Differentiate between periodic and non-periodic motion.

Answer: Periodic motion repeats after fixed intervals of time, while non-periodic motion does not repeat at regular intervals.

Q4. What is the difference between oscillatory and vibratory motion?

Answer: Both involve to-and-fro motion. Vibratory motion is a rapid form of to-and-fro motion.

Q5. Can a motion be both periodic and oscillatory?

Answer: Yes. The motion of a simple swinging pendulum is an example of periodic oscillatory motion.

Q6. Give an example of an object showing linear and rotational motion together.

Answer: A moving bicycle shows linear motion as it moves forward, while its wheels rotate.

Q7. What type of motion is shown by a child on a swing?

Answer: Oscillatory motion.

Q8. What type of motion is shown by the blades of a running fan?

Answer: Rotational motion.

Q9. What type of motion is shown by a car moving along a straight road?

Answer: Linear or rectilinear motion.

Q10. What type of motion is shown by a bee flying irregularly around flowers?

Answer: Random motion.

14. Important Multiple-Choice Questions (MCQs)

Practice Quiz

0/10 answered

1. What is the SI unit of length?

A. Kilometre
B. Metre
C. Centimetre
D. Foot

2. Which is a non-standard unit of length?

A. Metre
B. Centimetre
C. Handspan
D. Millimetre

3. Which tool is suitable for measuring a curved line?

A. Thread
B. Brick
C. Stone
D. Clock

4. A reference point is used to describe the ____ of an object.

A. Colour
B. Taste
C. Position
D. Weight only

5. Which motion occurs along a straight line?

A. Random

B. Rectilinear
C. Oscillatory
D. Vibratory

6. A rotating fan shows which type of motion?

A. Rotational
B. Only linear
C. Only random
D. Non-periodic only

7. Sound is produced by ____.

A. Colour
B. Vibrations
C. Light
D. Weight

8. Which motion repeats after fixed intervals of time?

A. Random motion
B. Non-periodic motion
C. Periodic motion
D. Irregular motion

9. 1 kilometre is equal to ____ metres.

A. 10
B. 100
C. 1000
D. 10,000

10. Which mode of transport includes aeroplanes?

A. Land
B. Water
C. Air
D. Rail only
Check Answers

15. Fill in the Blanks

1. The SI unit of length is the metre.
2. A handspan is a non-standard unit of measurement.
3. The distance between two places can be measured in metres or kilometres.

4. The motion of an object along a straight line is called rectilinear motion.
5. A thread can be used to measure the length of a curved line.
6. The motion of a rotating fan is called rotational motion.
7. Sound is produced by vibrations.
8. Motion that repeats after fixed intervals is called periodic motion.
9. A fixed object used to describe another object's position is called a reference point.
10. The invention of the wheel made transport easier.

16. Very Short-Answer Questions

1. Name the SI unit of length. — Metre.
2. Name one standard measuring tool. — Ruler.
3. Name one non-standard unit. — Handspan.
4. What is motion? — A change in position with time.
5. What is rest? — No change in position with time relative to a reference point.
6. What is the motion of a swinging pendulum? — Oscillatory motion.
7. What is the motion of a fan? — Rotational motion.
8. What is the motion of a car on a straight road? — Rectilinear motion.
9. What is the SI unit of time? — Second.
10. What is the SI unit of mass? — Kilogram.

17. Important Differences for Exams

Pair	Difference
Standard and non-standard units	Standard units have fixed values; non-standard units may differ from person to person.
Motion and rest	In motion, position changes with time; at rest, position does not change relative to a reference point.
Linear and curvilinear motion	Linear motion follows a straight path; curvilinear motion follows a curved path.
Rotational and oscillatory motion	Rotational motion involves turning around an axis; oscillatory motion involves repeated to-and-fro movement.
Periodic and non-periodic motion	Periodic motion repeats at fixed intervals; non-periodic motion does not repeat at regular intervals.
Oscillatory and vibratory motion	Vibratory motion is rapid to-and-fro motion; oscillatory motion is the broader to-and-fro movement.

18. Quick Revision: Key Facts

- The three modes of transport are land, water and air.
- The wheel was an important invention in the development of transport.
- Measurement tells us the size or amount of a quantity.

- The SI unit of length is metre (m).
- $1\text{ km}=1000\text{ m}$.
- $1\text{ m}=100\text{ cm}$.
- $1\text{ cm}=10\text{ mm}$.
- A thread helps measure a curved line.
- A reference point helps describe an object's position.
- Motion may be rectilinear, curvilinear, rotational, oscillatory, vibratory, periodic or non-periodic.
- A single object can show more than one type of motion at the same time.

Note: The uploaded PDF contains the chapter's teaching material and activities, but no separate end-of-chapter exercise page is visible in the 10 pages provided. The answers above cover the questions and activities visible in the file, along with additional topic-wise practice questions.